DataForge AI - Master Product Requirements Document (PRD)

Version: 1.0 (Hackathon Freeze)

Vision

Build an autonomous AI Data Engineering Platform that discovers, evaluates, curates, improves, generates only when necessary, validates, explains and exports production-ready datasets optimized for any foundation model.

Problem

AI dataset engineering is fragmented across many tools for search, cleaning, licensing, quality assessment, synthetic generation, formatting and export.

Core Philosophy

Search → Evaluate → Curate → Merge → Clean → Translate → Balance → Generate (only if required) → Benchmark → Explain → Export.

Target Users

AI Engineers, ML Engineers, Data Scientists, Research Labs, Enterprises, Startups, Universities.

Key Features

- Requirement Analyzer
- Dynamic Question Generator
- Planner / Orchestrator
- Dynamic Execution Graph
- Dataset Discovery
- License Validation
- Quality Evaluation
- Cleaning & Deduplication
- Merge Agent
- Translation Agent
- Synthetic Generation Agent
- Bias Detection
- Critic
- Validator
- Benchmark Agent
- Formatter
- Packaging
- Explainability (Auditor)
- Dataset Intelligence Report

• Hybrid API Router
• Bring Your Own Model (BYOM)

Workflow

User Request → Requirement Analyzer → Planner → Execution Graph → Dataset Discovery → Curation → Evaluation → If insufficient: Synthetic Generation → Critic → Validator → Benchmark → Quality Gate → Formatter → Packaging → Export

Quality Profiles

Fast, Balanced, Production, Research, Enterprise.

Hybrid Router

Supports NVIDIA NIM, OpenAI, Gemini, Anthropic, Groq, Together AI, HuggingFace, Ollama and Custom OpenAI-compatible APIs. Automatic failover based on quota, latency, outage or policy.

NVIDIA Stack

Primary:
- NVIDIA NIM
- NVIDIA Nemotron
- NeMo Agent Toolkit
- NeMo Curator
- NeMo Retriever
- Triton Inference Server
- TensorRT-LLM
- CUDA

Roadmap:
- NVIDIA AI Foundry

Explainability

Every important decision is logged: dataset rejection, license conflicts, routing changes, synthetic generation trigger, quality iterations and stopping reason.

Dataset Intelligence Report

Includes:
- Executive Summary
- Dataset Overview
- Composition
- Language Distribution
- Quality Metrics
- Bias Analysis
- Coverage Analysis
- Training Readiness
- Model Compatibility

- Cost Report
- Decision Timeline
- Recommendations
- Certification

Deliverables

dataset.zip
- dataset
- metadata
- dataset card
- quality report
- explainability report
- benchmark report
- bias report
- license report
- statistics
- dataset intelligence report
- recommendations

Tech Stack

Frontend: Next.js, React, Tailwind, shadcn/ui
Backend: FastAPI, Python
Agent Framework: NeMo Agent Toolkit + LangGraph
Storage: PostgreSQL, Redis
Vector DB: Milvus/Qdrant/FAISS

MVP

Implement Requirement Analyzer, Planner, Dynamic Graph, Discovery, Curator, Quality Evaluation, License Checker, Synthetic Generator, Critic, Validator, Explainability Agent, Hybrid Router, Formatter, Dataset Intelligence Report and ZIP export.

Future Roadmap

Multi-modal support, Agent Marketplace, Dataset Digital Twin, Feedback Learning, Enterprise SaaS, Team Workspaces, Plugins.

Product Statement

DataForge AI is an autonomous AI Data Engineer that discovers, evaluates, curates, improves, generates only when necessary, validates, explains and delivers production-ready datasets through dynamic multi-agent orchestration.

DataForge AI – Master PRD (Continuation)

Section 16 – System Architecture

16. System Architecture

DataForge AI follows a modular, event-driven, multi-agent architecture.

Instead of executing a fixed workflow, the system dynamically creates an execution graph based on the user's requirements.

```
                    USER
                     │
                     ▼
             Requirement Analyzer
                     │
                     ▼
             Clarification Generator
                     │
                     ▼
               Planner (Brain)
                     │
          Creates Dynamic Execution Graph
                     │
┌────────────────────┬────────────────────────┬────────────────────────┐
▼         ▼          ▼          ▼
Discovery  Curator   Generator   Validator
│         │          │          │
└──────────────────────┴──────────┬──────────┴─────────────────────────┘
                ▼
        Quality Evaluation
                │
```

```
   Quality Threshold Reached?
        │         │
       YES        NO
        │         │
        ▼         │
    Formatter        │
        │         │
        ▼         │
  Packaging & Reports ◄──────┘
        │
        ▼
      Export
```

---

17. Core Components

17.1 User Interface

Responsibilities

- Receive user requirements
- Show execution progress
- Visualize workflow
- Display reports
- Download outputs

17.2 Requirement Engine

Purpose

Convert natural language into structured project requirements.

Example

User says

I need 50k multilingual legal conversations for Nemotron.

Output

Purpose:

Fine-tuning

Domain:

Legal

Languages:

English

Hindi

Quality:

Production

Output:

JSONL

Target Model:

Nemotron

17.3 Planner (Brain)

The Planner is the central intelligence.

Responsibilities

- Understand requirements

- Design workflow
- Select agents
- Estimate cost
- Estimate execution time
- Estimate quality
- Generate execution graph

The Planner NEVER directly edits datasets.

It only coordinates work.

17.4 Dynamic Execution Graph

Every request creates a unique graph.

Example A

Search

↓

Curator

↓

Formatter

Example B

Search

↓

Merge

↓

Translate

↓

Benchmark

↓

Formatter

Example C

Synthetic Generation

↓

Validator

↓

Critic

↓

Benchmark

↓

Formatter

No workflow is hardcoded.

---

18. Agent Architecture

Every agent is independent.

Agents communicate through structured messages.

Planner can spawn only the required agents.

18.1 Requirement Analyzer

Purpose

Convert user goals into machine-readable tasks.

Inputs

Natural language

Outputs

Structured requirements

18.2 Clarification Agent

Purpose

Ask only necessary questions.

Examples

Need language?

Need dataset size?

Need commercial license?

Need model compatibility?

Questions are adaptive.

18.3 Planner Agent

Responsibilities

- Task decomposition
- Workflow generation
- Agent allocation
- Cost prediction
- Quality prediction

Planner decides

Should we

Search?

Generate?

Translate?

Merge?

Reject?

18.4 Dataset Discovery Agent

Responsibilities

Search

- HuggingFace
- Kaggle
- GitHub
- Research datasets

- Government datasets
- Enterprise repositories

Returns

Candidate datasets

Metadata

License

Size

Quality

18.5 License Agent

Checks

Commercial use

Academic only

Research only

Open Source

License conflicts

Rejected datasets include explanation.

18.6 Curator Agent

Uses NVIDIA NeMo Curator.

Responsibilities

Deduplicate

Normalize

Remove noisy data

PII Detection

Semantic filtering

Language filtering

Format correction

18.7 Merge Agent

Responsibilities

Merge multiple datasets

Resolve schema conflicts

Remove overlaps

Maintain consistency

18.8 Cleaning Agent

Correct

Broken records

Missing values

Invalid formats

Encoding issues

18.9 Translation Agent

Supports

Single language

Multi-language

Code-switching

Regional languages

Maintains semantic consistency.

18.10 Synthetic Generation Agent

Generation occurs ONLY when

Planner determines existing data is insufficient.

Uses

Nemotron

or

User selected model.

Generates

Missing samples

Edge cases

Rare scenarios

Long-tail examples

18.11 Bias Detection Agent

Evaluates

Gender

Age

Location

Language

Domain

Topic distribution

Representation

Produces Bias Report.

18.12 Critic Agent

Challenges generated outputs.

Checks

Hallucinations

Consistency

Naturalness

Prompt leakage

Formatting

Requests regeneration if needed.

18.13 Validator Agent

Validates

Completeness

Accuracy

Consistency

Quality

Readiness

18.14 Benchmark Agent

Scores

Quality

Coverage

Bias

Duplicates

Readability

Training readiness

Outputs Benchmark Report.

18.15 Explainability Agent

The Auditor.

Records every decision.

Examples

Why dataset rejected

Why API switched

Why synthetic generation occurred

Why iteration repeated

Why quality threshold passed

Produces

Decision Trail

Explainability Report

18.16 Formatter Agent

Converts outputs into

JSON

JSONL

CSV

Parquet

ZIP

Custom schemas

18.17 Packaging Agent

Packages

Dataset

Metadata

Reports

Documentation

Statistics

Into one downloadable archive.

---

19. Agent Communication Protocol

Agents communicate using structured JSON.

Example

{

"agent":"QualityEvaluator",

"dataset_id":"HF-1023",

"score":91.6,

"status":"PASS",

"recommendation":"Proceed to Formatter"

}

All agent outputs are versioned.

Every message is logged.

---

20. Workflow Engine

Execution States

WAITING

↓

PLANNING

↓

DISCOVERY

↓

CURATION

↓

GENERATION

↓

VALIDATION

↓

BENCHMARK

↓

FORMATTING

↓

PACKAGING

↓

COMPLETED

---

21. State Manager

Maintains

Current execution

Completed tasks

Running agents

Failures

Retries

Quality history

Routing history

---

22. Retry Strategy

If an agent fails

↓

Retry

↓

Different model

↓

Different provider

↓

Alternative workflow

↓

Planner notified

↓

Continue execution

---

23. Quality Gate

Each Quality Profile defines thresholds.

Example

Fast

Minimum 70%

Balanced

80%

Production

90%

Research

95%

Enterprise

98%

If quality is below threshold

↓

Planner starts another refinement cycle.

---

24. Stopping Conditions

Execution stops when

Quality threshold achieved

OR

Maximum iterations reached

OR

Budget exceeded

OR

User timeout

OR

User cancellation

Explainability Agent records the reason.

---

25. Cost Intelligence Engine

Planner estimates

Estimated API Cost

Estimated GPU Usage

Estimated Execution Time

Estimated Quality

Example

Estimated Cost

$2.14

Estimated Time

8 minutes

Expected Quality

94%

Recommendation

Balanced Profile

Planner optimizes

Quality

Latency

Cost

Together.

End of Part 2

Next continuation (Part 3) will cover:

- Database Schema
- API Specifications
- Complete UI/UX
- Hybrid API Router (deep design)
- NVIDIA SDK Integration
- Dataset Intelligence Report (full specification)

- Security
- Performance
- Deployment Architecture
- Monitoring
- Logging
- MVP Milestones
- Development Roadmap
- Competitive Analysis
- Final Product Positioning

DataForge AI – Master PRD (Continuation)

Part 3 – Database, APIs, UI/UX, NVIDIA Integration, Deployment

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26. Database Architecture

The platform uses a hybrid storage architecture.

Relational Database (PostgreSQL)

Stores

- Users
- Projects
- API Keys (Encrypted)
- Execution History
- Workflow Definitions
- Reports Metadata
- Audit Logs

Cache (Redis)

Stores

- Active Sessions
- Running Agent State
- Planner Memory
- Execution Queue
- Temporary Results

Vector Database (Milvus / Qdrant)

Stores

- Dataset Embeddings
- Metadata Embeddings
- Semantic Search Index
- Agent Memory
- Similar Dataset Retrieval

Object Storage

Stores

- Dataset ZIPs
- Reports
- Generated Images
- Logs
- Intermediate Files

Database Schema

Users

- User ID

- Name
- Email
- Subscription
- Created At

Projects

- Project ID
- User ID
- Project Name
- Status
- Quality Profile
- Target Model

Workflows

- Workflow ID
- Planner Output
- Agent Graph
- Execution Status
- Cost
- Time

Datasets

- Dataset ID
- Source
- License
- Size

- Languages
- Domain
- Quality Score

Reports

- Report ID
- Project ID
- Type
- Generated Time
- Download URL

Audit Logs

- Decision
- Agent
- Timestamp
- Reason
- Confidence

---

27. API Architecture

REST + WebSocket

Authentication

POST /login

POST /register

POST /refresh

Projects

POST /project/create

GET /project/{id}

DELETE /project/{id}

Workflow

POST /workflow/start

GET /workflow/status

GET /workflow/graph

Agents

GET /agents

POST /agents/run

GET /agents/status

Reports

GET /reports

GET /reports/download

Dataset

POST /dataset/build

POST /dataset/search

POST /dataset/export

Hybrid Router

POST /router/provider

POST /router/test

GET /router/status

WebSocket Events

Planner Started

Agent Spawned

Dataset Found

Generation Started

Quality Updated

API Switched

Benchmark Complete

Packaging Complete

Download Ready

---

28. User Interface

Dashboard

Displays

Recent Projects

Execution History

Saved APIs

Quality Profiles

Usage Statistics

Create Dataset

Large prompt box

Example

Build a multilingual finance dataset for fine-tuning Nemotron.

Smart Form

Generated dynamically.

Questions appear only when required.

Example

Language?

Domain?

Commercial use?

Target Model?

Quality Profile?

Output Format?

Workflow Viewer

Real-time graph.

Shows

Planner

↓

Discovery

↓

Curator

↓

Generation

↓

Validator

↓

Formatter

Each node updates live.

Agent Monitor

Shows

Running Agents

Completed Agents

Current Step

Execution Time

Token Usage

API Used

Dataset Explorer

Preview dataset.

Search.

Filter.

Statistics.

Reports Page

View

Quality Report

Bias Report

Explainability Report

Benchmark

Dataset Intelligence Report

Settings

Manage

API Keys

Providers

Models

Quality Profiles

Themes

Notifications

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29. Hybrid API Router

Purpose

Route every agent to the most suitable model.

Supported Providers

NVIDIA NIM

OpenAI

Anthropic

Gemini

Groq

Together AI

Ollama

HuggingFace

Custom

Routing Policies

Fastest

Cheapest

Highest Quality

Balanced

Privacy

Custom Priority

Smart Failover

If

Quota exceeded

↓

Switch Provider

If

Timeout

↓

Switch Provider

If

Poor Quality

↓

Switch Model

If

Unavailable

↓

Planner chooses backup.

Every switch recorded by Auditor.

---

30. NVIDIA Integration

NVIDIA NIM

Used by

Planner

Requirement Analyzer

Generator

Critic

Validator

Explainability

NVIDIA Nemotron

Primary reasoning model.

Handles

Planning

Critique

Validation

Dataset Generation

Quality Evaluation

NeMo Agent Toolkit

Agent lifecycle

Execution graph

Communication

State management

NeMo Curator

Dataset cleaning

Semantic deduplication

PII removal

Filtering

Quality improvement

NeMo Retriever

Semantic search

Dataset retrieval

Metadata search

Knowledge retrieval

Triton Inference Server

Model serving

Batch inference

Scalable execution

TensorRT-LLM

Low latency inference

GPU optimization

CUDA

Acceleration

Embeddings

Evaluation

Preprocessing

Future

AI Foundry

DGX Deployment

Distributed Training

---

31. Dataset Intelligence Report

Every exported dataset contains

Executive Summary

Dataset Name

Purpose

Status

Overall Quality

Dataset Overview

Samples

Languages

Domains

Token Statistics

Compression

Composition

Category Distribution

Topic Distribution

Conversation Types

Coverage Analysis

Covered Topics

Missing Topics

Weak Areas

Suggested Improvements

Language Analysis

Language %

Code-switch %

Vocabulary Richness

Quality Analysis

Duplicates

Noise

Missing Fields

Formatting

Consistency

Overall Score

Bias Analysis

Gender

Age

Region

Language

Toxicity

Representation

Training Readiness

Recommended

SFT

RAG

DPO

Evaluation

RLHF

Compatibility

Nemotron

Llama

Gemma

Qwen

DeepSeek

GPT

Execution Summary

Execution Time

API Calls

Providers Used

Synthetic Ratio

Real Data Ratio

Decision Timeline

Complete Planner history.

Improvement Suggestions

What should be improved next.

Certification

Research Ready

Production Ready

Enterprise Ready

---

32. Explainability Report

Human-readable.

Explains

Why datasets rejected

Why merged

Why generated

Why translated

Why router switched

Why iterations occurred

Why execution stopped

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33. Security

Encrypted API Keys

Role-based Access

JWT Authentication

HTTPS

Rate Limiting

Audit Logging

Secret Management

PII Detection

License Validation

---

34. Performance Targets

Planner

<2 sec

Agent Spawn

<1 sec

Search

<10 sec

Quality Evaluation

<20 sec

Generation

Variable

UI Updates

Realtime

---

35. Monitoring

Track

Latency

Failures

Retries

API Cost

GPU Usage

Quality

Provider Health

Planner Decisions

---

36. Logging

Every event logged.

Every decision logged.

Every retry logged.

Every provider switch logged.

Every quality iteration logged.

---

37. Error Recovery

Automatic Retry

Fallback Provider

Fallback Model

Alternative Workflow

Planner Recovery

User Notification

---

38. MVP

The hackathon MVP includes:

✓ Requirement Analyzer

✓ Dynamic Question Generator

✓ Planner

✓ Dynamic Execution Graph

✓ Dataset Discovery

✓ NeMo Curator

✓ Quality Evaluation

✓ License Validation

✓ Synthetic Generation

✓ Critic

✓ Validator

✓ Explainability

✓ Hybrid Router

✓ Formatter

✓ Packaging

✓ Dataset Intelligence Report

✓ ZIP Export

End of Part 3

Final Part 4 will include:

- Development Timeline
- Competitive Analysis
- Why This Wins Track C
- NVIDIA Presentation Mapping
- Risks & Mitigations
- Future SaaS Architecture
- Product Roadmap
- Success Metrics

- Final Product Positioning
- Complete Conclusion

This will complete the **master PRD**. It will be one coherent document assembled from these continuations.

39. Functional Requirements

FR-001 Requirement Collection

The system shall accept natural language dataset requests.

FR-002 Requirement Clarification

The system shall dynamically ask only the minimum required clarification questions.

FR-003 Dynamic Planning

The Planner shall create a unique execution graph for every request.

FR-004 Agent Orchestration

The Planner shall instantiate only the agents required for the current workflow.

FR-005 Dataset Discovery

The system shall search multiple public and private dataset repositories.

FR-006 Dataset Evaluation

The system shall score candidate datasets based on quality, relevance, completeness and licensing.

FR-007 License Validation

The system shall reject incompatible datasets based on user requirements.

FR-008 Dataset Curation

The system shall clean, normalize, deduplicate and improve datasets before use.

FR-009 Synthetic Generation

Synthetic data shall only be generated when existing datasets cannot satisfy the requested quality.

FR-010 Hybrid Routing

The system shall support multiple inference providers and automatic failover.

FR-011 Explainability

Every major planner and agent decision shall be recorded and explained.

FR-012 Export

The system shall export datasets and all reports as a downloadable package.

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40. Non-Functional Requirements

Performance

Planner response <2 seconds

Workflow initialization <5 seconds

Real-time UI updates

Reliability

Automatic retries

Graceful degradation

Provider failover

Scalability

Support thousands of concurrent dataset generation requests.

Security

Encrypted API keys

Role-based access

Secure communications

Maintainability

Modular agent architecture

Independent agents

Reusable workflows

---

41. User Personas

AI Engineer

Needs production-ready datasets quickly.

ML Engineer

Needs datasets optimized for model training.

Researcher

Needs transparent, explainable datasets.

Enterprise

Needs compliant, licensed, production-quality datasets.

Student

Needs datasets for experiments and projects.

---

42. Success Metrics (KPIs)

Dataset Quality Score

Dataset Coverage

Bias Score

Duplicate Ratio

Execution Time

Planner Accuracy

Average Iterations

Generation Percentage

Synthetic-to-Real Ratio

Average API Cost

User Satisfaction

Successful Dataset Completion Rate

---

43. Competitive Analysis

Feature	Hugging Face	Gretel	Mostly AI	Scale AI	DataForge AI
Dataset Search	✅	❌	❌	❌	✅
Cleaning	Partial	✅	✅	Partial	✅
License Validation	❌	❌	❌	❌	✅
Dynamic Planning	❌	❌	❌	❌	✅
Multi-Agent Workflow	❌	❌	❌	❌	✅
Explainability	❌	Partial	Partial	❌	✅
Dataset Intelligence Report	❌	❌	❌	❌	✅
Hybrid Model Router	❌	❌	❌	❌	✅

---

44. Assumptions

- Internet connectivity is available.
- Dataset repositories expose searchable metadata.
- Users possess valid API keys when using external providers.
- GPU resources are available for inference.
- Public datasets provide sufficient metadata for evaluation.

---

45. Constraints

Hackathon duration

API quotas

GPU memory

Execution time

Budget

Public dataset licensing

Network availability

---

46. Risk Analysis

Risk	**Impact**	**Mitigation**
API Failure	High	Hybrid Router
Poor Dataset Quality	High	Curator + Validator
License Conflict	High	License Agent
Hallucination	Medium	Critic + Validator
Budget Exhaustion	Medium	Cost Optimizer
Long Execution Time	Medium	Dynamic Workflow Optimization

---

47. Testing Strategy

Unit Testing

Individual agents

Planner

Router

Reports

Integration Testing

Planner

↓

Agents

↓

Reports

System Testing

Complete workflow

Performance Testing

Large datasets

Concurrent requests

Stress Testing

Maximum workload

User Acceptance Testing

Real AI engineers

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48. Deployment Architecture

```
          User
           │
        Next.js Frontend
           │
        FastAPI Backend
           │
      Planner / Orchestrator
           │
   ┌──────────────┼──────────────┐
   │       │       │
Agent Pool   Hybrid Router   State Manager
   │       │       │
   │    NVIDIA NIM / APIs   │
```

```
      |                  |
 PostgreSQL   Redis   Vector DB   Object Storage
```

---

49. NVIDIA Technology Justification

NVIDIA NIM

Provides production-grade inference endpoints for agent execution.

Nemotron

Used for reasoning-intensive tasks such as planning, validation and critique.

NeMo Agent Toolkit

Provides structured multi-agent orchestration.

NeMo Curator

Reduces engineering effort by providing industrial-grade dataset cleaning, deduplication and filtering.

NeMo Retriever

Provides semantic retrieval across datasets.

Triton

Efficient inference serving.

TensorRT-LLM

Optimized GPU inference.

---

50. GitHub Repository Structure

dataforge-ai/

frontend/

backend/

planner/

agents/

router/

reports/

database/

deployment/

config/

tests/

docs/

README.md

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51. Development Timeline

Week 1

Core architecture

Planner

Requirement Analyzer

UI

Week 2

Dataset Discovery

NeMo Curator

Hybrid Router

Week 3

Synthetic Generation

Critic

Validator

Reports

Week 4

Testing

Optimization

Demo

Presentation

Documentation

---

52. Demo Flow

Step 1

User enters:

"Create a multilingual healthcare dataset for Nemotron."

Step 2

Planner generates execution graph.

Step 3

Dataset search begins.

Step 4

Curator cleans datasets.

Step 5

Synthetic generation fills missing gaps.

Step 6

Validator approves.

Step 7

Dataset Intelligence Report generated.

Step 8

User downloads complete ZIP package.

---

53. Why DataForge AI Wins Track C

Unlike traditional synthetic data generators, DataForge AI autonomously performs the complete AI data engineering lifecycle.

It:

- Searches before generating.
- Validates before exporting.
- Explains every decision.
- Optimizes quality, cost and latency.
- Produces production-ready datasets instead of raw synthetic outputs.

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54. Future Roadmap

Phase 1

Hackathon MVP

Phase 2

Enterprise deployment

Phase 3

Dataset Marketplace

Phase 4

Agent Marketplace

Phase 5

Dataset Digital Twin

Phase 6

Cloud SaaS Platform

Phase 7

Self-improving AI Data Engineer

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55. Final Product Statement

DataForge AI is an autonomous AI Data Engineering Platform that intelligently discovers, evaluates, curates, improves, generates only when necessary, validates, explains and delivers production-ready datasets through dynamic multi-agent orchestration powered by NVIDIA's AI ecosystem.